Relative Discriminant Charges and Signed Source Response in Theta Heat Flow

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Abstract

We develop a finite-cluster framework for studying multiple zeros of the backward heat evolution of the Riemann xi function. The local degree of the holomorphic jet (H, H_z) equals the winding of the cluster discriminant; an event of spatial multiplicity m has charge $\binom{m}{2}$. For a proper two-root cluster, the discriminant satisfies a regular affine equation $\Delta' = 8 + A\Delta$. A complex-time phase bound reduces its charge to the signs at two real time caps. We construct explicit complete-source windows at every hypothetical finite-height double, derive a signed discriminant response with a quadratic remainder, and control event displacement along an affine heat homotopy. The canonical theta tail has a positive Laplace-mixture representation that strictly improves a matched pointwise-envelope enclosure, without yet determining the required sign. For arbitrary finite clusters we prove $D' = 8T + BD$ and exhibit exact cubic and quartic obstructions to a direct extension of the two-root cap rule. Existing interval certificates and exact arithmetic checks provide supporting examples. The missing theta-specific signed estimate, its height coverage, and the higher multiplicity exclusion remain explicit open obligations. No endpoint exclusion or RH-conditional simplicity theorem is claimed.

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1 The endpoint problem and the complete source

The purpose of this paper is to assemble the analytic results of the relative-charge investigation into a single argument with explicit hypotheses. The question motivating the investigation is whether the unchanged theta source supplies enough signed information to exclude a multiple zero at time zero, conditional on the Riemann hypothesis (RH). The results below supply geometric windows, exact response identities, and quantitative estimates to which such information could be applied. They also identify limitations of several simpler substitutes.

1.1 Normalization and analytic bounds

Throughout the theta application, put

$$H(t,z) = \int_0^\infty \Phi(u) e^{tu^2} \cos(zu)\, du, \tag{1.1}$$

$$\Phi(u) = \sum_{n\ge 1} \phi_n(u), \qquad \lambda_n = \pi n^2,$$

$$\phi_n(u) = (2\lambda_n^2 e^{9u} - 3\lambda_n e^{5u}) e^{-\lambda_n e^{4u}}. \tag{1.2}$$

This is the standard normalization used in [1]:

$$H(0,z) = \frac{1}{8}\xi\left(\frac{1}{2} + \frac{iz}{2}\right), \qquad \xi(s) = \frac{s(s-1)}{2}\pi^{-s/2}\Gamma(s/2)\zeta(s). \tag{1.3}$$

The Fourier representation (1.3) and the classical real-zero threshold theorem are the analytic number-theory inputs; none of the local proofs below uses a zero table or RH unless it is explicitly stated.

For $u \ge 0$ every summand in (1.2) is positive, since $2\lambda_n e^{4u} - 3 > 0$. The series and its derivatives converge uniformly on compact real u intervals. For large u the summands are bounded by a polynomial in n, e^u times $e^{-\pi n^2 e^{4u}}$; factoring out, for example, $e^{-(\pi/2)e^{4u}}$ leaves a uniformly summable series. Consequently Φ and each of its fixed-order derivatives decay faster than e^{-cu^2-du} for any fixed real c, d. Differentiation under the integral is therefore justified locally uniformly in $(t,z) \in \mathbb{C}^2$. In particular,

$$H_t = -H_{zz}, \qquad H(t,-z) = H(t,z), \qquad H(\bar t, \bar z) = \overline{H(t,z)}. \tag{1.4}$$

Here and below subscripts denote derivatives; the time parameter is written as an argument to avoid confusing it with a derivative.

For $T \in \mathbb{R}$, $R \ge 0$ and nonnegative integer j, define the *complete moment*

$$M_j(T,R) = \int_0^\infty \Phi(u) u^j e^{Tu^2} \cosh(Ru)\, du. \tag{1.5}$$

Since $|\cos(x+iy)|$ and $|\sin(x+iy)|$ are at most $\cosh y$,

$$|\partial_t^a \partial_z^b H(t,z)| \le M_{2a+b}(T,R) \quad (\operatorname{Re} t \le T,\ |\operatorname{Im} z| \le R). \tag{1.6}$$

These bounds are independent of $\operatorname{Re} z$. "Complete" means that every source mode and the entire integration tail are included. For a signed residual source ψ, the same assertion holds with $|\psi|$ in place of Φ.

1.2 What RH supplies, and what it leaves open

The de Bruijn–Newman theorem gives a finite real Λ for which all zeros of $H(t,\cdot)$ are real exactly when $t \geq \Lambda$. In this normalization RH is equivalent to $\Lambda \leq 0$; Rodgers and Tao proved $\Lambda \geq 0$ [2]. Thus under RH the zeros are real for all $t \geq 0$. They are simple for $t > 0$: if a real zero had multiplicity $m \geq 2$ at $t_0 > 0$, the local Hermite splitting proved in Proposition 2.4, applied at $t = t_0 - \varepsilon$, would give nonreal zeros for sufficiently small $\varepsilon > 0$. At least two Hermite roots are nonzero, and their imaginary displacement has order $\sqrt{\varepsilon}$, whereas the error is $O(\varepsilon)$. This contradicts reality at that time. The argument does not rule out a multiple zero at $t = 0$.

A common negative time at which *all* theta zeros are real is unavailable, even under RH, since it would contradict $\Lambda \geq 0$. This restriction does not prohibit different negative-time neighborhoods for different endpoint heights. The quantitative windows below are explicitly allowed to depend on the hypothetical event.

The local charge at an endpoint of multiplicity m will be $q = \binom{m}{2}$. It relates directly to the multiplicity excess studied in the companion heat-motion investigation [15, 16]. Under RH let $\rho = \frac{1}{2} + i\gamma$ range over distinct nontrivial zeros of both ordinate signs, with multiplicity m_ρ. For every finite symmetric cutoff Y, define

$$K_Y = \sum_{|\gamma| \leq Y} \frac{m_\rho^2}{|\rho|^2}, \qquad Z_Y = \sum_{|\gamma| \leq Y} \frac{m_\rho}{|\rho|^2}. \tag{1.7}$$

Termwise algebra gives

$$K_Y - Z_Y = 2 \sum_{|\gamma| \leq Y} \frac{q_\rho}{|\rho|^2}. \tag{1.8}$$

When the two weighted sums converge, as in the companion framework, the same identity holds without the cutoff. No convergence theorem is needed for the local arguments in this paper. At the physical coordinate $a = 2\gamma$, the charge weight on the right is $8/(1 + a^2)$. Thus the local integer charge does not produce a positive uniform gap for the weighted excess. An endpoint-local charge must also be distinguished from the charge of a larger window, which might contain a negative-time event.

1.3 Degree, relative Chern number, and Hopf interpretation

Let $\Omega \subset \mathbb{C}^2$ be a bounded domain with piecewise smooth boundary, and suppose $G = (H, H_z)$ is nonzero on that boundary. Equip both copies of $\mathbb{C}^2$ with the complex orientation, in the coordinate order $(\operatorname{Re} t, \operatorname{Im} t, \operatorname{Re} z, \operatorname{Im} z)$. The boundary charge is

$$Q_\Omega = \deg\left(\frac{G}{\|G\|} : \partial\Omega \longrightarrow S^3\right). \tag{1.9}$$

It is the relative top Chern number of the trivial complex rank-two bundle with the specified nonvanishing boundary section: equivalently, the total index of its interior zeros. For isolated holomorphic zeros these indices are positive analytic intersection multiplicities. Theorem 2.3 supplies the discriminant computation of them.

For the product of a time rectangle and a spatial disk, rounding corners makes the boundary an oriented S^3. Its usual Hopf interpretation requires no additional charge formula. Let $\pi : S^3 \to S^2$ be the Hopf map, let ω have integral one on S^2, and choose a connection form α with $d\alpha = \pi^*\omega$ and $\int_{S^3} \alpha \wedge d\alpha = 1$. For $g = G/\|G\|$ the Hopf invariant of $\pi \circ g$ is

$$\int_{\partial\Omega} g^*\alpha \wedge d(g^*\alpha) = \int_{\partial\Omega} g^*(\alpha \wedge d\alpha) = \deg g.$$

Here the integral formula defines the normalized Hopf invariant; its independence of the chosen primitive follows by Stokes' theorem and $H^1(S^3) = 0$. It agrees with (1.9). The analytic estimates in this paper use degree and discriminants directly.

Neither reality nor spatial parity supplies cancellation. Conjugating both complex domain coordinates preserves real orientation, as does conjugating both target coordinates. The change $z \mapsto -z$ and its target action $(H, H_z) \mapsto (H, -H_z)$ are also orientation preserving. The corresponding reflected events therefore have the same positive local charge.

1.4 Reading the results

The first part proves proper cluster preparation, local charge, the two-root cap theorem and explicit event-compatible windows. The second part retains signs in source perturbations, proves quantitative event continuation, and specializes the response to the actual canonical theta tail. The many-root part keeps every root of a higher birth and shows exactly where the two-root argument ceases to apply. Numerical evidence is collected separately in Section 9. Appendix A records the holomorphic reference comparison and the boundary interpolation contract used by earlier certificates.

All propositions are statements with the hypotheses displayed here. In particular, a narrower source-response interval need not lie on one side of zero, and a phase bound need not imply zero charge. This manuscript consolidates internally reviewed research results; it has not received independent specialist validation. It asserts no priority for the standard complex-analysis and heat-root-motion ingredients.

2 Finite clusters and their positive charge

The objects in this section are finite clusters, rather than a product over all zeros of an entire function. This distinction makes the boundary assumptions explicit and retains the influence of the exterior zeros through a holomorphic unit. The construction consolidates [6].

Definition 2.1 (Proper spatial cluster)**.** Let $D \subset \mathbb{C}$ be a bounded disk or rectangle, and let $\Omega_R = \{z : |z - a| < R\}$. Suppose H is holomorphic on a neighborhood of $\overline{D} \times \overline{\Omega}_R$ and

$$H(t, z) \neq 0 \qquad (t \in \overline{D},\ |z - a| = R).$$

The zeros of $H(t, \cdot)$ in Ω_R, counted with multiplicity, form a proper cluster. Their number m is independent of t. A collar of radius $r < R$ means that all these zeros lie in $\Omega_r = \{|z - a| < r\}$ throughout $\overline{D}$.

The count is the integer-valued continuous function $(2\pi i)^{-1} \oint H_z/H\, dz$. In particular, two counts, one on each of the circles of radii r and R, certify a collar; an inner count alone does not exclude additional roots in the annulus.

Lemma 2.2 (Preparation without root labels)**.** *A proper cluster has a monic polynomial*

$$P(t, z) = z^m + p_1(t)z^{m-1} + \cdots + p_m(t)$$

with coefficients holomorphic in t, and $H(t, z) = U(t, z)P(t, z)$, where U is holomorphic and nonvanishing on the product. The polynomial is uniquely determined by the cluster.

Proof. For $k \geq 1$ the contour integrals

$$S_k(t) = \frac{1}{2\pi i} \oint_{|\zeta - a| = R} \zeta^k \frac{H_z(t, \zeta)}{H(t, \zeta)}\, d\zeta \tag{2.1}$$

are holomorphic and equal the sums of the kth powers of the roots, including multiplicity. Newton's identities determine the elementary symmetric functions recursively from $S_1, \dots, S_m$, hence construct P with holomorphic coefficients. On each fixed time fiber H/P has removable singularities at exactly the selected roots and has no zeros in the disk. Joint holomorphy follows, without choosing those roots, from the Cauchy representation

$$U(t,z) = \frac{1}{2\pi i} \oint_{|\zeta - a| = R} \frac{H(t,\zeta)}{P(t,\zeta)(\zeta - z)}\, d\zeta, \qquad |z-a| < R.$$

For each t this is the removable quotient, and the integrand is holomorphic in its parameters. Near the spatial wall the quotient itself supplies the extension. Compactness and the strict boundary condition allow the same construction on a slightly enlarged time neighborhood. Uniqueness follows from the multiset of roots. □

Define the finite discriminant by

$$\Delta_P(t) = \operatorname{Disc}_z P(t,z) = \prod_{i<j} (z_i(t) - z_j(t))^2. \tag{2.2}$$

The symmetric expression is holomorphic even when the roots cannot be labeled holomorphically. We set $\Delta_P = 1$ if $m \le 1$.

Theorem 2.3 (Boundary charge and discriminant winding)**.** *In addition to properness, suppose that (H, H_z) has no zero on $\partial D \times \overline{\Omega}_R$. Give complex two-space its complex orientation and the product boundary its induced orientation. The degree of*

$$\frac{(H, H_z)}{\sqrt{|H|^2 + |H_z|^2}} : \partial(D \times \Omega_R) \longrightarrow S^3$$

is well defined and equals

$$Q_{D \times \Omega_R} = \frac{1}{2\pi i} \oint_{\partial D} \frac{\Delta_P'(t)}{\Delta_P(t)}\, dt. \tag{2.3}$$

It is the sum of the local complex intersection multiplicities of (H, H_z), and is therefore a nonnegative integer.

Proof. The side wall contains no zero of H, and the remaining boundary contains no joint zero by hypothesis. The discriminant is nonzero on ∂D. It is consequently not identically zero and has only finitely many zeros in $\overline{D}$. Above each such time there are only finitely many cluster roots, so the joint zeros are isolated.

Fix one such time t_0 and put $\mathcal{R} = \mathbb{C}\{t - t_0\}$, the ring of convergent power series. The module $\mathcal{A} = \mathcal{R}[z]/(P)$ is free of rank m over $\mathcal{R}$. Multiplication by P_z on $\mathcal{A}$ has determinant

$$\operatorname{Res}_z(P, P_z) = (-1)^{m(m-1)/2} \Delta_P.$$

Indeed, at a time with distinct roots multiplication is diagonal in the evaluation coordinates, with determinant $\prod_i P_z(z_i)$; the identity then extends holomorphically. The ring $\mathcal{R}$ is a discrete valuation ring. Elementary row and column reduction puts any matrix with nonzero determinant into diagonal form with entries units times powers $(t - t_0)^{e_j}$. Thus the order of its determinant is the sum of the e_j, which is also the length of its cokernel. Applying this to multiplication by P_z gives

$$\operatorname{ord}_{t_0} \Delta_P = \operatorname{length}_{\mathcal{R}} \mathcal{A}/(P_z).$$

Small disjoint spatial disks around the distinct roots at t_0 split this quotient into its local factors. In each factor, analytic division by the monic local polynomial identifies its length with

$$\dim_{\mathbb{C}} \frac{\mathbb{C}\{t-t_0, z-b\}}{(P, P_z)}.$$

Since U is a unit, $(H, H_z) = (UP, UP_z + U_z P)$ and (P, P_z) generate the same ideal. These dimensions are therefore precisely the local intersection multiplicities of the original jet.

For completeness, the topological interpretation uses the usual local degree property of a holomorphic square map: at an isolated zero its degree is the dimension of its local analytic quotient. One way to see the sign and multiplicity is to perturb the target to a nearby regular value in an isolating neighborhood. The local intersection number counts the preimages with their multiplicities; at a regular preimage the real Jacobian determinant is $|\det_{\mathbb{C}} DF|^2 > 0$. Conservation of the intersection number and homotopy invariance of boundary degree give the asserted equality at the original value. Additivity over disjoint isolating neighborhoods then identifies the product-boundary degree with the sum of the above lengths. The scalar argument principle sums $\operatorname{ord}_{t_0} \Delta_P$ and proves (2.3). Corners of a product rectangle may be rounded within a boundary neighborhood free of joint zeros, without changing either degree. □

The heat equation determines the local multiplicity in this formula. Here and below the time normalization is $H_t = -H_{zz}$.

Proposition 2.4 (Hermite birth and local charge)**.** *Suppose $H(t_0, b+w) = c_m w^m + O(w^{m+1})$, where $c_m \neq 0$ and $m \geq 2$. The event (t_0, b) is an isolated zero of (H, H_z). For its local m-root cluster, writing $s = t - t_0$ and $N = m(m-1)/2$, one has*

$$\Delta_P(t_0 + s) = d_m s^N + O(s^{N+1}), \qquad d_m = 2^N \prod_{i<j} (\lambda_i - \lambda_j)^2 > 0, \tag{2.4}$$

where $\lambda_1, \dots, \lambda_m$ are the roots of the probabilists' Hermite polynomial He_m*. The local jet charge is N.*

Proof. The heat equation recursively determines the Taylor coefficients in s from the initial spatial Taylor series. Terms arising from w^ℓ have weighted degree ℓ when w has weight one and s has weight two. The leading weighted polynomial is

$$c_m e^{-s\partial_w^2} w^m = c_m (2s)^{m/2} \,\mathrm{He}_m\left(\frac{w}{\sqrt{2s}}\right).$$

This is a polynomial identity, so the notation involving square roots introduces no choice into the polynomial itself.

The polynomial He_m has m distinct real roots. Indeed, its Rodrigues formula and integration by parts imply orthogonality to every polynomial of degree less than m for the positive weight $e^{-x^2/2}$. If it had fewer than m distinct real roots of odd multiplicity, multiply it by the product of the linear factors at just those roots. The resulting real polynomial would have a constant nonzero sign except at finitely many points, so its weighted integral could not vanish. Orthogonality contradicts this. Degree m now forces all m roots to be real and simple.

Set $s = \varepsilon^2$ and $w = \varepsilon x$. Taylor convergence gives, uniformly for bounded x,

$$\varepsilon^{-m} H(t_0 + \varepsilon^2, b + \varepsilon x) = c_m 2^{m/2} \,\mathrm{He}_m(x/\sqrt{2}) + O(\varepsilon).$$

The left side extends holomorphically to $\varepsilon = 0$. The simple-root implicit function theorem gives all m cluster roots as $w_i = \varepsilon(\sqrt{2}\lambda_i + O(\varepsilon))$. They are distinct for sufficiently small nonzero ε. Their

squared differences have product $d_m\varepsilon^{2N}(1+O(\varepsilon))$. The discriminant is holomorphic in $s=\varepsilon^2$, so its Taylor series has only even powers of ε; this proves (2.4). It also proves local isolation of the event. Theorem 2.3 identifies its charge with the order N. □

Consequently, on any product satisfying Theorem 2.3 for a heat family,

$$Q_{D\times\Omega_R} = \sum_{(t,b):\, H(t,b)=H_z(t,b)=0} \binom{m(t,b)}{2}. \tag{2.5}$$

Interior event charges do not cancel. Reality and spatial parity do not change this: simultaneous conjugation in two complex coordinates has real orientation sign $+1$, and spatial sign reversal is complex linear, also of sign $+1$. These symmetries can pair equal charges, but cannot force their sum to vanish.

3 The two-root equation and the real-cap theorem

Assume a proper cluster consists of two roots in a collar $0<r<R$. Write $H=UP$, $P=(z-z_1)(z-z_2)$, and $\Delta=(z_1-z_2)^2$. Set $q=U_z/U$ and, on the outer circle, $f=H_z/H$. The exterior unit U is retained throughout; replacing it by a constant would change the equation for the original cluster.

Proposition 3.1 (Regular affine discriminant flow)**.** *The holomorphic coefficient*

$$A(t)=\frac{4}{2\pi i}\oint_{|\zeta-a|=R}\frac{H_z(t,\zeta)}{H(t,\zeta)P(t,\zeta)}\,d\zeta \tag{3.1}$$

is regular through collisions and satisfies

$$\Delta' = 8 + A\Delta. \tag{3.2}$$

For any holomorphic scalar $\kappa(t)$, H_z may be replaced in (3.1) by $H_z-\kappa H$. If on the outer wall $|H|\ge\delta>0$ and $|H_z-\kappa H|\le L_\kappa$, then

$$|A|\le K := \frac{4RL_\kappa}{\delta(R-r)^2}. \tag{3.3}$$

Every zero of Δ is simple and has derivative 8.

Proof. At a simple root the heat equation gives $z_i'=H_{zz}/H_z$. Differentiating UP at that root yields

$$z_1' = \frac{2}{z_1-z_2}+2q(t,z_1), \qquad z_2' = \frac{2}{z_2-z_1}+2q(t,z_2).$$

Thus

$$\Delta' = 8+4\Delta\frac{q(t,z_1)-q(t,z_2)}{z_1-z_2}.$$

By residues the divided difference is $(2\pi i)^{-1}\oint q/P\,d\zeta$. Since $f=q+P_z/P$ and $P_z/P^2=-\partial_z(1/P)$, the integral is unchanged if q is replaced by f. This proves the equation away from events. The discriminant is not identically zero by Proposition 2.4; alternatively, a persistent double $H=U(z-c(t))^2$ would give $H_t(t,c(t))=0$ and $H_{zz}(t,c(t))=2U(t,c(t))\neq 0$, contradicting heat flow. Both sides of (3.2) are holomorphic, so the equation extends through its isolated events.

The scalar subtraction is exact because $\oint 1/P\,d\zeta=0$: the rational function $1/P$ has no exterior finite poles and decays quadratically at infinity. On the outer wall, $|P|\ge(R-r)^2$. Estimating the contour length $2\pi R$ gives (3.3). Substitution of $\Delta=0$ in the equation gives $\Delta'=8$. □

There is no inverse internal root separation in (3.1) or (3.3). The bound does pay for the exterior collar and the spatial-wall floor. A scalar centering κ improves it only if the corresponding coupled numerator is actually estimated more sharply.

Theorem 3.2 (Reduction of complex charge to two real cap signs)**.** *Let*

$$D = \{x + iy : t_- < x < t_+,\ |y| < \tau\}, \qquad t_- < t_+,$$

and suppose the proper two-root and collar hypotheses hold on a neighborhood of its closure. Let $a \in \mathbb{R}$ *and assume reality* $H(\bar{t}, \bar{z}) = \overline{H(t, z)}$*. It suffices that*

$$\sup_{x+iy \in \overline{D}} \left| \int_0^y \operatorname{Re} A(x + is)\, ds \right| < \frac{\pi}{2}. \tag{3.4}$$

In particular, $|A| \le K$ *and* $K\tau < \pi/2$ *suffice. Then there is at most one cluster event in* $\overline{D}$*, and any such event has real time and real spatial position. If* $\Delta(t_-)\Delta(t_+) \neq 0$*, the entire product boundary is free of joint zeros, and*

$$Q_{D \times \Omega_R} = \frac{\operatorname{sgn} \Delta(t_+) - \operatorname{sgn} \Delta(t_-)}{2} \in \{0, 1\}. \tag{3.5}$$

Proof. Choose $t_* \in (t_-, t_+)$ and define the holomorphic primitives

$$J(t) = \int_{t_*}^t A(v)\, dv, \qquad \mu(t) = e^{-J(t)}, \qquad \Psi(t) = \int_{t_*}^t \mu(v)\, dv.$$

They are single valued on the rectangle. Reality and uniqueness of the prepared polynomial imply reality of A on the real interval. Integrating first horizontally and then vertically gives

$$\operatorname{Im} J(x + iy) = \int_0^y \operatorname{Re} A(x + is)\, ds.$$

Therefore (3.4) implies $\operatorname{Re} \mu > 0$ on the closed rectangle. Convexity gives, for distinct s, t in its closure,

$$\frac{\Psi(t) - \Psi(s)}{t - s} = \int_0^1 \mu(s + v(t - s))\, dv,$$

whose real part is positive. Thus Ψ is injective, including on the boundary. Multiplying (3.2) by μ and integrating yields

$$\mu(t)\Delta(t) = \Delta(t_*) + 8\Psi(t). \tag{3.6}$$

There is at most one zero. A nonreal zero would have a distinct conjugate zero, so any zero is real. At a real event the real monic quadratic has a repeated root, which is necessarily real as well.

The only possible event times on the boundary are now t_- and t_+, and the cap assumptions exclude them. On real time $\mu > 0$ and $(\mu\Delta)' = 8\mu > 0$. Equal nonzero cap signs give no event; negative then positive signs give exactly one. The reversed sign pattern cannot occur. Each event contributes one by Proposition 3.1 and Theorem 2.3, proving (3.5). □

The theorem proves an upper bound of one event, not zero events. The heat family $H = z^2 - 2t$ has $A = 0$ and $\Delta = 8t$; on $|\operatorname{Re} t|, |\operatorname{Im} t| \le 1/4$ its roots lie in $|z| < 1$, so $r = 1, R = 2$ supply a proper collar and charge one. The phase condition is entirely compatible with this collision. This distinction is central to the endpoint problem [7].

3.1 A complete-source strip criterion

For the complete positive source, use the moments

$$M_j(T,s)=\int_0^\infty \Phi(u)u^j e^{Tu^2}\cosh(su)\,du.$$

They are finite, and no source modes or exterior tail are omitted. Suppose $d_r, d_R>0$ are established bounds

$$|H(x,z)|\ge d_s \quad (t_-\le x\le t_+,\ |z-a|=s), \qquad s=r,R,$$

and a real anchor has root count two on both circles. Put $B_s=M_2(t_+,s)$ and $L=M_1(t_+,R)$. Then a sufficient width is

$$0<\tau<\min\left\{\frac{d_r}{B_r},\frac{d_R}{B_R+8RL/(\pi(R-r)^2)}\right\}. \tag{3.7}$$

Indeed, $|e^{iyu^2}-1|\le |y|u^2$ and $|\cos(zu)|,|\sin(zu)|\le\cosh(su)$ on a circle centered at a real a give

$$|H(x+iy,z)-H(x,z)|\le \tau B_s, \qquad |H_z(x+iy,z)|\le M_1(t_+,s).$$

Both walls therefore have positive floors $d_s-\tau B_s$, and continuity of their counts gives the proper two-root collar throughout the strip. Applying (3.3) with $\kappa=0$ and $\delta=d_R-\tau B_R$ gives $K\tau<\pi/2$ exactly from the second inequality in (3.7). Purely imaginary time lifting requires no extra factor $e^{\tau u^2}$. The cap signs still require separate information. The finite verified application discussed later uses this type of complete-source payment [8].

4 Explicit windows at a hypothetical double

The preceding theorem has a pointwise geometric contract at every finite real height. This section pays it in terms of the local second jet, including when the two roots coincide [9]. Fix $T_0,R_0>0$ and set

$$B_j=M_j(T_0,R_0), \qquad j=2,3. \tag{4.1}$$

The complete source gives $|\partial_z^j H(t,z)|\le B_j$ whenever $\operatorname{Re} t\le T_0$ and $|\operatorname{Im} z|\le R_0$. These bounds do not depend on $\operatorname{Re} z$.

Theorem 4.1 (Quantitative double window)**.** *Assume, conditionally, that for a real a,*

$$H(0,a)=H_z(0,a)=0, \qquad c=\tfrac12 H_{zz}(0,a)\neq 0, \qquad q_0=|c|.$$

Define

$$\begin{aligned} R&=\min\{R_0/2,q_0/B_3\}, \qquad r=R/2,\\ h&=\min\left\{T_0/2,\frac{q_0R^2}{16\sqrt{2}B_2},\frac{R^2}{128}\right\}. \end{aligned} \tag{4.2}$$

For every t with $|\operatorname{Re} t|,|\operatorname{Im} t|\le h$, $H(t,\cdot)$ has exactly two roots inside each of the spatial circles of radii r and R, and their walls satisfy

$$|H(t,z)|>\tfrac12 q_0 s^2 \quad (|z-a|=s,\ s=r,R). \tag{4.3}$$

On the outer wall $|H_z|<3q_0R$. The regular coefficient satisfies

$$|A|\le K:=96/R^2, \qquad Kh\le 3/4<\pi/2. \tag{4.4}$$

The hypotheses extend to a neighborhood of the closed product.

Proof. For $|w| = s \le R$, spatial Taylor expansion with integral remainder and straight-line integration in complex time give

$$|H(0, a + w) - cw^2| \le B_3 s^3/6 \le q_0 s^2/6,$$

$$|H(t, a + w) - H(0, a + w)| \le |t| B_2 \le \sqrt{2} h B_2 \le q_0 R^2/16 = q_0 r^2/4.$$

At $s = r, R$ their sum is at most $5q_0 s^2/12 < q_0 s^2$. Rouché's theorem against cw^2 gives both counts and the stronger common floor $7q_0 s^2/12$. At the outer circle the same calculation gives $37q_0 R^2/48$. Counting twice excludes annular roots.

For the first derivative use its spatial remainder and $H_{zt} = -H_{zzz}$:

$$\begin{aligned} |H_z(t, a + w)| &\le 2q_0 R + \tfrac{1}{2} B_3 R^2 + \sqrt{2} h B_3 \\ &\le \left(\tfrac{5}{2} + \frac{\sqrt{2}}{128} \right) q_0 R < 3 q_0 R. \end{aligned}$$

The last step uses $RB_3 \le q_0$ and $h \le R^2/128$. Equation (3.3), with $\delta = q_0 R^2/2$ and $R - r = R/2$, now gives $K = 96/R^2$. All required wall margins are strict and $Kh \le 3/4 < \pi/2$; continuity on the compact boundary preserves the conditions on a small enlargement. The corner factor $\sqrt{2}$ was included explicitly in the time estimates. □

Corollary 4.2 (The compatible cap signs)**.** *In Theorem 4.1 define, for $x \ge 0$,*

$$L_K(x) = \frac{8}{K}(1 - e^{-Kx}), \qquad U_K(x) = \frac{8}{K}(e^{Kx} - 1), \tag{4.5}$$

with continuous limiting values $8x$ if $K = 0$. Then

$$\begin{aligned} L_K(h) \le \Delta(h) \le U_K(h), \\ -U_K(h) \le \Delta(-h) \le -L_K(h). \end{aligned} \tag{4.6}$$

The resulting product has charge one, with its unique event at $(0, a)$.

Proof. At the assumed double, $\Delta(0) = 0$. On real time the affine equation has the exact solution

$$\Delta(t) = 8 \int_0^t \exp\left(\int_s^t A(v)\, dv \right) ds.$$

The coefficient is real and bounded by K, so comparison with $e^{-K|t-s|}$ and $e^{K|t-s|}$ gives (4.6). In particular the cap margins are positive, and $L_K(h) \ge 8he^{-Kh}$. Theorem 3.2 gives the charge and uniqueness. □

This is a geometry theorem conditional on an exact double. It cannot also be an exclusion theorem: its actual cap signs are those of a collision. Improving an absolute coefficient bound changes their quantitative margins, but cannot turn them into equal signs.

4.1 A finite-jet test away from an assumed event

At any real center let $c_0 = H(0, a)$, $c_1 = H_z(0, a)$, $c_2 = H_{zz}(0, a)/2$, and assume $q_0 = |c_2| > 0$. Choose $0 < r < R \le R_0$ and $\sqrt{2}h \le T_0$. If, for $s = r, R$,

$$d_s = q_0 s^2 - |c_0| - s|c_1| - B_3 s^3/6 - \sqrt{2} h B_2 > 0, \tag{4.7}$$

then the same Rouché proof gives two roots inside each circle and wall floors d_s on the closed time square. Moreover

$$L = |c_1| + 2q_0R + B_3R^2/2 + \sqrt{2}hB_3, \qquad K = \frac{4RL}{d_R(R-r)^2} \tag{4.8}$$

are valid outer derivative and coefficient bounds. If $Kh < \pi/2$, the real-cap theorem applies once the two cap discriminants are enclosed away from zero. The test does not assert an event, an internal gap, or either cap sign. An interval implementation needs a lower enclosure for q_0 and complete upper enclosures for every error term; uncertain jet equalities cannot be promoted to an exact double.

4.2 Event duration and the height quantifier

If an event is already known to occur at real time $s < t_*$ and $|A| \le K$ between them, put $d = t_* - s$ and $\delta = \Delta(t_*) > 0$. The preceding integral solution gives $L_K(d) \le \delta \le U_K(d)$. Therefore, if $0 < \delta_- \le \delta \le \delta_+$ and $K\delta_+ < 8$,

$$\frac{\log(1 + K\delta_-/8)}{K} \le d \le -\frac{\log(1 - K\delta_+/8)}{K}. \tag{4.9}$$

For $K = 0$ the interval is $[\delta_-/8, \delta_+/8]$. These inequalities require event existence, supplied for example by opposite cap signs; a single positive cap value does not supply it. The scalar bounds are sharp under the information $|A| \le K$, as constant coefficients $A = K$ and $A = -K$ attain the two extremes. This statement does not assert that either coefficient is realized by theta. A strictly negative event location would follow only from data making the resulting upper bound for s negative.

The moments in (4.1) work for every finite real a. Thus every hypothetical exact double has a positive window from (4.2), point by point. No uniform lower bound for its second jet is proved. In fact, with $f(u) = u^2\Phi(u)$ and $C = \|f'\|_{L^1(0,\infty)}/2 < \infty$, integration by parts gives, for $a \neq 0$,

$$q_0(a) = \frac{1}{2}\left|\int_0^\infty f(u)\cos(au)\,du\right| \le \frac{C}{|a|}.$$

Consequently this particular prescription obeys

$$R(a) \le \frac{C}{B_3|a|}, \qquad h(a) \le \frac{C^3}{16\sqrt{2}B_2B_3^2|a|^3}. \tag{4.10}$$

These are upper bounds for conservatively chosen window sizes, not necessary scales for every possible construction. On a bounded real interval there are only finitely many zeros of the nonzero entire function $H(0,\cdot)$, so the windows of its exact doubles there have positive minima if the set is nonempty. This does not compute those minima or identify any doubles. An all-height exclusion may work pointwise and need not use a common negative cap, but its signed estimate must cover every admissible finite a and the associated shrinking parameters. The geometric quantifier is paid here; that signed quantifier remains open.

5 Signed source response and displacement of a double event

The two-root flow separates the analytic geometry of a possible collision from the source information needed to exclude it. This section gives a signed source functional, an explicit nonlinear error, and a continuation criterion for the event itself. The results consolidate the analytic arguments in [10]; their proofs are included here.

5.1 An exact response formula at a fixed real time

Fix a real time, which is suppressed in the notation, and a real spatial center a. Let F and $H = F+E$ be holomorphic near $\overline{B(a,R)}$. Assume that F is nonzero on $\Gamma = \{z : |z-a| = R\}$, has exactly two zeros counted with multiplicity inside Γ, and that both zeros belong to $B(a,r)$, where $0 < r < R$. Write c for their centroid and Δ_F for their squared difference. The reference zeros may coincide. Throughout this section contours have counterclockwise orientation, and

$$\mathcal{I}[g] = \frac{1}{2\pi i}\int_\Gamma g(z)\,dz.$$

Suppose the complete wall estimate is

$$\rho := \sup_\Gamma |E/F| < 1. \tag{5.1}$$

Rouché's theorem then gives two zeros of $F + sE$ inside Γ for $0 \le s \le 1$. A count in a smaller disk, if needed for the two-root flow theorem, is an additional condition; it does not follow from (5.1) alone.

Theorem 5.1 (Logarithmic discriminant response)**.** *Let $\ell = \log(1 + E/F)$ be the branch defined by its convergent power series on an annulus containing Γ. Then*

$$\Delta_H - \Delta_F = -4\mathcal{I}[(z-c)\ell] - \big(\mathcal{I}[\ell]\big)^2. \tag{5.2}$$

Set

$$\mathcal{L}_F(E) = -4\mathcal{I}[(z-c)E/F], \tag{5.3}$$

$$l_1(\rho) = -\log(1-\rho), \qquad l_2(\rho) = -\log(1-\rho) - \rho,$$

$$\mathcal{B}(\rho) = 4R(R+r)l_2(\rho) + R^2 l_1(\rho)^2. \tag{5.4}$$

The finite response satisfies

$$|\Delta_H - \Delta_F - \mathcal{L}_F(E)| \le \mathcal{B}(\rho). \tag{5.5}$$

In particular the error is $O(\rho^2)$, with constants independent of the distance between the two reference zeros.

Proof. Put $w = z - a$, and for $G = F, H$ let $S_k(G) = \mathcal{I}[w^k G_z/G]$ be the sum of the kth powers of its two centered zeros, counting multiplicity. On the annulus, $H_z/H - F_z/F = \ell_z$. Integration by parts on the closed contour gives

$$S_1(H) - S_1(F) = -\mathcal{I}[\ell], \qquad S_2(H) - S_2(F) = -2\mathcal{I}[w\ell].$$

The identity $\Delta_G = 2S_2(G) - S_1(G)^2$, together with $S_1(F) = 2(c-a)$, proves (5.2). No extension of ℓ across the zeros of F was used. Its power series yields $|\ell| \le l_1(\rho)$ and $|\ell - E/F| \le l_2(\rho)$ on Γ. Now $|\mathcal{I}[g]| \le R\sup_\Gamma |g|$ and $|z-c| \le R + r$ prove (5.5). □

If F and H are real-symmetric, their discriminants and $\mathcal{L}_F(E)$ are real. Thus a signed enclosure $\mathcal{L}_F(E) \in [L_-, L_+]$ gives

$$\Delta_F + L_- - \mathcal{B}(\rho) \le \Delta_H \le \Delta_F + L_+ + \mathcal{B}(\rho). \tag{5.6}$$

A positive lower bound at a negative real cap can be combined with the two-root charge theorem once all its spatial and phase hypotheses have been established. A positive lower bound at $t = 0$ itself excludes an endpoint double in this fiber. Formula (5.6) does not provide either strict inequality without further source information. It also retains the external wall ratio and the radii; being finite at an internal double is not a uniform condition estimate over all heights.

5.2 The complete signed source functional

Suppose, now displaying the fixed real time t, that

$$E(t,z)=\int_0^\infty \psi(u)e^{tu^2}\cos(zu)\,du.$$

If $|F(t,z)|\ge\delta>0$ on Γ and

$$N_\psi(t,R):=\int_0^\infty |\psi(u)|e^{tu^2}\cosh(Ru)\,du<\infty, \tag{5.7}$$

then Fubini's theorem gives

$$\mathcal{L}_F(E)=\int_0^\infty \psi(u)e^{tu^2}K_F(u)\,du, \tag{5.8}$$

$$K_F(u)=-4\mathcal{I}\left[\frac{(z-c)\cos(zu)}{F(t,z)}\right],\qquad |K_F(u)|\le\frac{4R(R+r)}{\delta}\cosh(Ru). \tag{5.9}$$

Indeed $|\operatorname{Im} z|\le R$ on the circle because a is real, so $|\cos(zu)|\le\cosh(Ru)$. This proves an integrable majorant for the exchanged integrals, and also $\rho\le N_\psi(t,R)/\delta$. Conjugation symmetry makes K_F real on the positive real u axis.

At an exact double $b=c$, factor $F(t,z)=U(t,z)(z-b)^2$ with U a nonvanishing analytic function. A residue in (5.3) then gives the particularly simple identity

$$\mathcal{L}_F(E)=-\frac{4E(t,b)}{U(t,b)}=-\frac{8E(t,b)}{F_{zz}(t,b)}. \tag{5.10}$$

The sign is therefore an oscillatory source measurement divided by a nonzero curvature. Positivity of the source and absolute moment bounds alone do not determine it. For numerical use, every omitted source mode, unbounded integration ray, contour arc, and quadrature error must be included in both the wall ratio and the signed functional. No new numerical acquisition is used to prove this identity.

5.3 Differentiating a real event through an affine heat family

Let $H_s(t,z)=F(t,z)+sE(t,z)$ be real-symmetric, with both F and E satisfying the backward heat equation. Suppose F has an ordinary real double at (τ_0,b_0), so $F=F_z=0$ and $F_{zz}\neq0$ there.

Proposition 5.2 (Event derivatives and a one-sided finite error)**.** *There is a unique local real-analytic event branch $s\mapsto(\tau_s,b_s)$. At a point on this branch, set*

$$N=H_{s,zz}(\tau_s,b_s),\quad M=H_{s,zzz}(\tau_s,b_s),\quad P_4=H_{s,zzzz}(\tau_s,b_s).$$

With the residual derivatives evaluated at the same point,

$$\tau_s'=\frac{E}{N},\qquad b_s'=\frac{M\tau_s'-E_z}{N}, \tag{5.11}$$

$$\tau_s''=\frac{P_4}{N}(\tau_s')^2-\frac{2E_{zz}}{N}\tau_s'-(b_s')^2. \tag{5.12}$$

If the branch exists for $0\le s\le1$ and along it

$$|N|\ge\nu>0,\quad |P_4|\le C_4,\quad |E|\le E_0,\quad |E_{zz}|\le E_2,$$

then

$$\tau_1\le\tau_0+\frac{E(\tau_0,b_0)}{F_{zz}(\tau_0,b_0)}+\frac12\left(\frac{C_4E_0^2}{\nu^3}+\frac{2E_2E_0}{\nu^2}\right). \tag{5.13}$$

Proof. The Jacobian of $(H_s, H_{s,z})$ with respect to (t, z) at an event is

$$\begin{pmatrix} -N & 0 \\ -M & N \end{pmatrix},$$

whose determinant is $-N^2$. The analytic implicit-function theorem gives the local branch, and real symmetry gives its reality. One derivative of the two event equations gives $-N\tau_s' + E = 0$ and $-M\tau_s' + Nb_s' + E_z = 0$. For a second derivative of the first equation, use $H_{s,tt} = P_4$, $H_{s,tz} = -M$, and $E_t = -E_{zz}$ to obtain

$$0 = -N\tau_s'' + P_4(\tau_s')^2 - 2M\tau_s'b_s' + N(b_s')^2 - 2E_{zz}\tau_s' + 2E_zb_s'.$$

Substitution of $E_z = M\tau_s' - Nb_s'$ leaves (5.12). All terms are real. In particular $-(b_s')^2 \le 0$, whereas $|\tau_s'| \le E_0/\nu$, so

$$\tau_s'' \le \frac{C_4E_0^2}{\nu^3} + \frac{2E_2E_0}{\nu^2}.$$

Integrating twice on $[0,1]$ proves (5.13). □

This calculation also checks the signs in the discriminant formula. At a double, $\partial_s\Delta_s = -8E/N$ by (5.10), while $\partial_t\Delta_s = 8$ by the two-root heat equation. Differentiating $\Delta_s(\tau_s) = 0$ recovers $\tau_s' = E/N$.

5.4 Paying continuation and curvature from common walls

The local implicit-function theorem by itself does not continue an event to $s = 1$. The following conditions supply that missing step and the curvature floor in (5.13).

Proposition 5.3 (A complete conditional continuation criterion)**.** *Let $\mathcal{D}$ be a closed time rectangle with real caps $t_- < t_+$ and imaginary half-width τ. Suppose every $H_s = F + sE$, $0 \le s \le 1$, has exactly two zeros inside $B(a, r)$ throughout $\mathcal{D}$, none in the closed collar to Γ, a common outer wall floor $|H_s| \ge \delta > 0$, and a common strict phase condition from the two-root real-cap theorem. Assume*

$$\Delta_F(t_-) < 0 < \Delta_F(t_+), \qquad m_{\mathrm{cap}} = \min\{-\Delta_F(t_-), \Delta_F(t_+)\} > 0.$$

If the two outer cap circles satisfy $|E/F| \le \rho < 1$ and

$$C_{\mathrm{cap}} := 4R(R + r)l_1(\rho) + R^2l_1(\rho)^2 < m_{\mathrm{cap}}, \tag{5.14}$$

then a unique real event exists for every $s \in [0, 1]$, the events form one real-analytic branch, and

$$|H_{s,zz}(\tau_s, b_s)| \ge \frac{2\delta}{(R + r)^2}. \tag{5.15}$$

Proof. Apply (5.2) to sE at each cap. The bound $|\ell| \le l_1(\rho)$ gives $|\Delta_s(t_\pm) - \Delta_F(t_\pm)| \le C_{\mathrm{cap}}$. Thus both cap signs persist. The two-root real-cap theorem gives exactly one event for each s, located on the real time interval. It has spatial multiplicity two and $\partial_t\Delta_s = 8$. The local branches of Proposition 5.2 agree wherever their domains overlap by uniqueness. They extend along the full compact parameter interval: any limit of event points remains away from the time caps by the strict cap margins and from the spatial boundary by the common walls, and is again a double. The implicit-function theorem then extends the branch at that limit.

For the curvature, write $H_s = U_sP_s$ in the disk. On Γ, $|P_s| \le (R + r)^2$, hence $|U_s| \ge \delta/(R + r)^2$. Since U_s has no zeros, the maximum modulus principle applied to $1/U_s$ propagates this bound throughout the disk. At an event, $P_s = (z - b_s)^2$ and $H_{s,zz} = 2U_s$, proving (5.15). □

Here are sufficient reference data for all the common hypotheses. Suppose F has two zeros inside each of the disks of radii r, R throughout $\mathcal{D}$, with wall floors $\delta_{F,r}, \delta_{F,R}$, and

$$Q_v = \sup_{t\in\mathcal{D},\ |z-a|=v} |E/F| < 1 \quad (v = r, R).$$

Rouché's theorem supplies both counts for every H_s, with floors $\delta_v = (1 - Q_v)\delta_{F,v}$. Take $\delta = \delta_R$. If a complete outer-wall derivative bound is $|F_z| + |E_z| \le L$, the sufficient common phase inequality is

$$\frac{4RL}{\delta(R-r)^2}\,\tau < \frac{\pi}{2}. \tag{5.16}$$

A more precise common estimate for the regular coefficient A_s may replace this one. If the source densities are Φ_0 and ψ, the remaining bounds along the real event branch can be taken to be

$$\begin{aligned} E_j &= \int_0^\infty |\psi(u)| u^j e^{t_+ u^2}\, du \quad (j = 0, 2), \\ C_4 &= \int_0^\infty (|\Phi_0(u)| + |\psi(u)|) u^4 e^{t_+ u^2}\, du. \end{aligned} \tag{5.17}$$

The real event positions allow $|\cos(b_s u)| \le 1$ in these bounds. All source modes and integration rays are included.

Consequently (5.13), together with (5.14)–(5.17), is a complete conditional criterion for locating the unique event below zero. It does not supply a negative right-hand side for the unchanged theta source. Both the signed initial velocity and the external wall margins remain substantive inputs.

Example 5.4 (An exact nonlinear and continuation check)**.** For

$$H_s(t, z) = z^2 - 2t + s(bz + c),$$

one has

$$\Delta_s(t) = 8t + s^2b^2 - 4sc, \qquad \tau_s = \frac{sc}{2} - \frac{s^2b^2}{8}, \qquad b_s = -\frac{sb}{2}.$$

The initial signed response is $-4c$, while the finite correction is s^2b^2. The logarithmic integral alone in (5.2) contributes $2s^2b^2$; its last square is essential. The event equations give $\tau_0' = c/2$ and $\tau_s'' = -b^2/4$, as required.

The continuation inequalities can also be checked with strict rational margins. Set $r = 1$, $R = 2$, $|\operatorname{Re} t|, |\operatorname{Im} t| \le 1/4$, $b = 1/16$, and $c = \pm 1/32$. Since $2|t| \le \sqrt{2}/2 < 3/4$, the inner and outer reference wall floors exceed $1/4$ and $13/4$. Thus the relative errors are less than $3/8$ and $5/104$, respectively, and both counts persist. The outer floor of H_s exceeds $99/32$, so (5.15) permits $\nu = 11/16$. Here $A_s = 0$ exactly, and the reference cap discriminants are $-2, +2$. Since $l_1(5/104) \le 5/99$,

$$C_{\mathrm{cap}} \le \frac{120}{99} + \frac{100}{9801} < 2.$$

For $c = -1/32$, one has $P_4 = E_{zz} = 0$, so (5.13) proves $\tau_1 \le -1/64 < 0$; its exact value is $-1/64 - 1/2048$. For $c = +1/32$ the event is positive. The same geometric conditions accommodate both outcomes; the sign comes from the signed displacement.

6 What finite complete envelope moments cannot determine

The obstruction proved here concerns precisely the envelope moments used in absolute source estimates. It does not apply to an arbitrary finite list of observations, which could include the oscillatory measurement one wishes to determine.

Theorem 6.1 (Fixed envelope moments with both event directions)**.** *Fix any finite collection*

$$k_j(u) = u^{n_j} e^{T_j u^2} \cosh(Y_j u), \qquad n_j \in \mathbb{Z}_{\geq 0}, \quad T_j \in \mathbb{R}, \quad Y_j \geq 0. \tag{6.1}$$

There is a strictly positive smooth even source Φ_0, with super-Gaussian decay, and a real smooth perturbation ψ supported in a compact subinterval of $(0,\infty)$, such that for all sufficiently small positive and negative s:

1. *$\Phi_s = \Phi_0 + s\psi$ remains positive, with the same source near zero and outside a fixed compact interval;*
2. *every selected complete moment $\int_0^\infty \Phi_s(u) k_j(u)\, du$ is independent of s;*
3. *the associated heat family has a real ordinary-double branch with $\tau_0 = 0$ and $\tau_0' > 0$.*

In particular the double event lies on either side of zero for the respective signs of sufficiently small s.

Proof. Take $c_0 > 0$ and

$$\Phi_0(u) = c_0 \left(e^{-u^4} + \tfrac{1}{2} e^{-(u-4)^4} + \tfrac{1}{2} e^{-(u+4)^4} \right).$$

This is positive, even, and rapidly decaying. If $G(z) = \int_0^\infty e^{-u^4} \cos(zu)\, du$, translation of the whole-line Fourier integral gives

$$H_0(0,z) = c_0 G(z)(1 + \cos 4z).$$

At $b = \pi/4$ the last factor has an ordinary double. To see that $G(b) > 0$ without numerical evaluation, write $M_j = \int_0^\infty u^j e^{-u^4}\, du$. Integration of $(ue^{-u^4})'$ gives $M_4 = M_0/4$; Cauchy–Schwarz gives $M_2 \leq (M_0 M_4)^{1/2} = M_0/2$. Since $\cos x \geq 1 - x^2/2$,

$$G(b) \geq M_0 - \frac{b^2}{2} M_2 \geq M_0(1 - b^2/4) > 0.$$

Consequently $H_{0,zz}(0,b) = 16 c_0 G(b) > 0$.

Choose a bounded open interval $I \Subset (0,\infty)$ and a smooth nonnegative bump χ supported in I and positive on a subinterval. In the real inner product $\langle f, g \rangle_\chi = \int_0^\infty \chi f g$, let p be the orthogonal projection of $f(u) = \cos(bu)$ onto the span of the k_j; linearly dependent functions may first be discarded. Set $\psi = \chi(f - p)$. Then

$$\int_0^\infty \psi k_j = 0, \qquad \int_0^\infty \psi(u) \cos(bu)\, du = \int_0^\infty \chi(u)(f(u) - p(u))^2\, du > 0. \tag{6.2}$$

For the strict inequality, $\cos(bu)$ cannot belong to the indicated finite span on any interval. Such an equality would extend to the whole real axis by analyticity. Expanding each hyperbolic cosine and grouping equal exponential factors writes its proposed right-hand side as a finite sum of polynomial multiples of $e^{Tu^2 + Yu}$ with real T, Y. As $u \to +\infty$, eliminate the largest quadratic exponent $T > 0$,

and within it the largest linear exponent, since a nonzero polynomial times that factor cannot remain bounded. Repeating removes all growing terms, including those with $T = 0, Y > 0$. All terms with $T < 0$, or with $T = 0, Y < 0$, decay. The remaining term is a polynomial, which must be constant to remain bounded. The resulting right-hand side has a limit, whereas $\cos(bu)$ does not. This contradiction proves (6.2).

The base source has a positive minimum on $\operatorname{supp}\psi$; therefore both signs of sufficiently small s preserve positivity. The even extension of ψ is smooth because its support is away from zero, and all prescribed moments remain exactly fixed. The super-Gaussian tail and the compact perturbation give jointly entire heat transforms. Proposition 5.2 and (6.2), divided by the positive curvature, give $\tau_0' > 0$. The local implicit-function theorem supplies the branch and the asserted signs. □

A sufficiently small proper two-root time-space window exists around the base double: isolate the zero at time zero, use nonvanishing on nested spatial circles to preserve the two counts, and shrink the time rectangle until the regular coefficient satisfies the strict phase condition. Its strict margins persist for all sufficiently small $|s|$ by uniform convergence on compact sets. Thus the two event directions coexist with the same selected moments and the same strict local analytic requirements.

Remark 6.2 (Scope of the control). The preserved moment vector is that of the constructed source Φ_0, not the theta source. Compact smooth perturbations need not preserve analyticity in the source variable, stronger shape conditions, or additional primitive identities. This construction asserts no global all-real heat slice, no all-positive-time real-rootedness, no canonical theta coefficient expansion, and no ordinary Dirichlet series or full normalized arithmetic reflection identity. It excludes only a universal inference from the displayed finite envelope moments and strict local analytic data. A measured signed functional such as (5.8), exact arithmetic data, and further consequences of RH lie outside that information class. In particular, this preservation list must not be combined with the distinct lists belonging to the other collision controls.

7 The canonical theta tail in the discriminant response

The actual theta tail has more structure than an arbitrary positive source bounded between two envelopes. We construct that structure explicitly, following the source identity in [13], and then apply it to the signed kernel. The resulting interval is strictly narrower than a specified pointwise-envelope interval, but is not proved to exclude zero.

7.1 An explicit positive Laplace law

Retain the complete mode normalization

$$\lambda_n = \pi n^2, \qquad \phi_n(u) = \lambda_n e^{5u}(2\lambda_n e^{4u} - 3)e^{-\lambda_n e^{4u}}.$$

Every mode is positive for $u \geq 0$. Fix an integer $m \geq 1$, and set

$$S_m = \sum_{n=1}^{m} \phi_n, \qquad R_m = \sum_{n>m} \phi_n,$$

$$\alpha = \lambda_m, \quad \beta = \lambda_{m+1}, \quad d = \beta - \alpha, \quad v = e^{4u} - 1, \quad D_0 = 2\alpha - 3, \quad \gamma = \frac{D_0}{2\alpha} > 0.$$

For $n > m$, define the nonnegative quantities

$$d_n = \lambda_n - \alpha, \qquad e_n = \lambda_n - \beta, \qquad a_n = \frac{\lambda_n^2}{\alpha^2} e^{-d_n}, \qquad b_n = \frac{3\lambda_n d_n}{\alpha^2 D_0} e^{-d_n}. \tag{7.1}$$

Proposition 7.1 (Canonical tail measure)**.** *Let* X_γ *have exponential density* $\gamma e^{-\gamma x}$ *on* $[0,\infty)$*, and put*

$$E_m = \sum_{n>m} (a_n + b_n) = \frac{R_m(0)}{\phi_m(0)},$$

$$\nu_m = \frac{1}{E_m} \sum_{n>m} \left[a_n \delta_{e_n} + b_n \operatorname{Law}(e_n + \mathsf{X}_\gamma) \right], \tag{7.2}$$

$$p_m(v) = \int_0^\infty e^{-xv}\, d\nu_m(x). \tag{7.3}$$

Then ν_m *is a probability measure and*

$$R_m(u) = E_m \phi_m(u) e^{-dv} p_m(v). \tag{7.4}$$

If $A = a_{m+1}$ *and* $\eta = A/E_m$*, then*

$$0 < \eta < 1, \qquad p_m(0) = 1, \qquad p_m(\infty) = \eta, \qquad (-1)^j p_m^{(j)}(v) \geq 0 \quad (j \geq 0, v \geq 0).$$

In particular

$$A\phi_m(u)e^{-dv} \leq R_m(u) \leq E_m \phi_m(u) e^{-dv}. \tag{7.5}$$

Proof. Division of the source polynomials, retaining their lower-order terms, gives

$$\frac{\phi_n(u)}{\phi_m(u)} = e^{-d_n v} \left(a_n + b_n \frac{\gamma}{\gamma + v} \right). \tag{7.6}$$

For example, with $q = 1 + v$ the needed algebra is

$$\frac{2\lambda_n q - 3}{2\alpha q - 3} = \frac{\lambda_n}{\alpha} + \frac{3(\lambda_n - \alpha)}{\alpha(2\alpha q - 3)}.$$

Also $\int_0^\infty e^{-xv} \gamma e^{-\gamma x}\, dx = \gamma/(\gamma + v)$. Factor e^{-dv} out of (7.6) and sum over all $n > m$ to obtain (7.4) and (7.2). Evaluating at $v = 0$ proves the expression for E_m and total mass one. The masses have Gaussian decay in n, and the continuous pieces have exponential tails, so every moment of ν_m is finite. Differentiation under its integral proves the alternating derivative inequalities, including at zero. Its only mass at zero is a_{m+1}/E_m; the exponential pieces have no atoms, and $b_{m+1} > 0$, so this number lies strictly between zero and one. Dominated convergence gives $p_m(\infty) = \eta$. The last enclosure follows. □

For later use, E_m/A is bounded as m varies and tends to one. Here is a direct estimate. Put $k = m + 1$ and

$$\varrho_m = \left(\frac{k+1}{k} \right)^4 e^{-\pi(2k+1)} < 1.$$

Successive ratios of a_n for $n \geq k$ are at most ϱ_m, and $b_n/a_n = 3d_n/(\lambda_n D_0) \leq 3/D_0$. Consequently

$$1 \leq \frac{E_m}{A} \leq \frac{1 + 3/D_0}{1 - \varrho_m} \longrightarrow 1. \tag{7.7}$$

This also supplies a complete elementary majorant for the infinite mass sum.

7.2 A cumulative-integral enclosure for the actual response

At a fixed real cap take the *unshifted* reference and residual

$$F(t,z)=\int_0^\infty S_m(u)e^{tu^2}\cos(zu)\,du,\qquad E(t,z)=\int_0^\infty R_m(u)e^{tu^2}\cos(zu)\,du. \tag{7.8}$$

Assume F has the proper two-root geometry of Section 5 and outer wall floor $\delta>0$. Define

$$\begin{aligned} W(u)&=e^{tu^2}\phi_m(u)e^{-d(e^{4u}-1)},\\ z_m(u)&=\frac{p_m(e^{4u}-1)-\eta}{1-\eta},\qquad d\sigma_m(w)=-z_m'(w)\,dw. \end{aligned} \tag{7.9}$$

The function z_m decreases from one to zero. Hence σ_m is a probability measure, and $z_m(u)=\sigma_m([u,\infty))$. The source identity becomes

$$e^{tu^2}R_m(u)=W(u)\{A+(E_m-A)z_m(u)\}. \tag{7.10}$$

For the real kernel in (5.9), put

$$J_K=\int_0^\infty W(u)K_F(u)\,du,\qquad C_K(w)=\int_0^w W(u)K_F(u)\,du,\qquad C_K(\infty)=J_K.$$

The bound in (5.9) and the complete source decay make these integrals absolutely convergent.

Theorem 7.2 (Canonical signed discriminant interval)**.** *With the preceding notation,*

$$\mathcal{L}_F(E)=AJ_K+(E_m-A)\int_0^\infty C_K(w)\,d\sigma_m(w). \tag{7.11}$$

Suppose a complete wall-ratio payment is

$$\overline{\rho}=\frac{E_m}{\delta}\int_0^\infty W(u)\cosh(Ru)\,du<1. \tag{7.12}$$

Then

$$\begin{aligned} \Delta_H&\ge\Delta_F+AJ_K+(E_m-A)\min_{0\le w\le\infty}C_K(w)-\mathcal{B}(\overline{\rho}),\\ \Delta_H&\le\Delta_F+AJ_K+(E_m-A)\max_{0\le w\le\infty}C_K(w)+\mathcal{B}(\overline{\rho}). \end{aligned} \tag{7.13}$$

The extrema exist on the compactified half-line. A sharper established wall ratio can replace (7.12)*.*

Proof. Substitute (7.10) into (5.8), write $z_m(u)=\int\mathbf{1}_{u\le w}\,d\sigma_m(w)$, and use Fubini. Absolute integrability follows from σ_m having mass one and $\int W|K_F|<\infty$. This proves (7.11). Continuity of C_K and its limit at infinity give a compact range, so its probability average lies between its minimum and maximum. The upper envelope in (7.5) proves $\sup_\Gamma|E/F|\le\overline{\rho}$. Finally (5.5), with the increasing function $\mathcal{B}$, proves (7.13). □

This enclosure retains every theta mode and the entire source ray. It is an analytic estimate, independent of a sampling scheme. An implementation must additionally enclose the acquisition errors and the full-domain extrema of C_K; no positive numerical lower endpoint is asserted here.

7.3 Strict improvement over a matched pointwise envelope

The following comparison keeps the same known lower and upper source bounds, the same wall ratio, and the same nonlinear response error. The pointwise comparison drops only the monotonicity of z_m.

Lemma 7.3 (Oscillation of the two-root response kernel)**.** *If both reference roots lie in* $\operatorname{Re} z > 0$*, then* K_F *has strictly positive and strictly negative intervals arbitrarily far along the positive* u *axis.*

Proof. Factor $F = UP$. For distinct roots z_1, z_2, their centroid satisfies $(z_i - c)/P_z(z_i) = 1/2$. The residue theorem therefore gives

$$K_F(u) = -2\left(\frac{\cos(z_1 u)}{U(z_1)} + \frac{\cos(z_2 u)}{U(z_2)}\right). \tag{7.14}$$

At a double $b > 0$ the formula is $K_F(u) = -4\cos(bu)/U(b)$, which has the asserted signs. For distinct real positive roots, (7.14) is a nonzero real combination of two distinct positive-frequency cosines. Its primitive is bounded and its mean square is positive: integrating the cosine products shows that the time average of the square tends to half the sum of the squared coefficients. If the kernel were eventually nonnegative, its integral on the remaining ray would be bounded, whereas $K_F^2 \le \|K_F\|_\infty K_F$ would then bound the square integral, contradicting its positive linear growth. Eventual nonpositivity is excluded in the same way. Continuity converts the resulting arbitrarily distant points of either sign into intervals.

The other possibility under real symmetry is a conjugate pair $b \pm iy$ with $b, y > 0$. Write $1/U(b + iy) = c + ie$. Equation (7.14) becomes

$$K_F(u) = -4\{c\cos(bu)\cosh(yu) + e\sin(bu)\sinh(yu)\}.$$

After division by e^{yu} this is a nonzero sinusoid plus a term tending exponentially to zero; $(c, e) \neq (0, 0)$ because U is nonvanishing. Its positive and negative lobes give the claim. □

Theorem 7.4 (Strict matched-envelope improvement)**.** *Under the hypotheses of Theorem 7.2, assume also that both reference roots lie in* $\operatorname{Re} z > 0$*. Set*

$$f(u) = W(u)K_F(u), \qquad P_+ = \int_0^\infty f_+(u)\,du, \qquad P_- = \int_0^\infty f_-(u)\,du,$$

where $f_+ = \max(f, 0)$ *and* $f_- = \max(-f, 0)$*. The pointwise relaxation using the same bounds* $AW \le e^{tu^2}R_m \le E_m W$ *has the exact interval for the linear response*

$$[AJ_K - (E_m - A)P_-,\ AJ_K + (E_m - A)P_+]. \tag{7.15}$$

There are explicit numbers $d_-, d_+ > 0$*, given by integrals over ordered sign intervals, for which*

$$\min C_K \ge -P_- + d_-, \qquad \max C_K \le P_+ - d_+. \tag{7.16}$$

Thus (7.13) *strictly improves both endpoints of the discriminant interval obtained from* (7.15) *with the identical nonlinear payment* $\mathcal{B}(\overline{\rho})$*.*

Proof. The pointwise relaxation permits an arbitrary measurable multiplier $0 \le a(u) \le 1$ in $W\{A + (E_m - A)a\}$. Choosing $a = 1$ exactly where f is positive or negative proves the two extrema in (7.15); intermediate values follow by convexity.

Since $W > 0$, Lemma 7.3 supplies a compact negative interval I_- followed by a compact positive interval I_+. Set

$$d_+ = \min\left\{\int_{I_-} f_-\,du, \int_{I_+} f_+\,du\right\} > 0.$$

For every w, the identity

$$P_+ - C_K(w) = \int_w^\infty f_+(u)\,du + \int_0^w f_-(u)\,du$$

is at least d_+: before the end of I_- the later positive interval is wholly omitted, and after that point the negative interval is wholly included. This proves the upper inequality. A positive interval followed by a negative one gives, by the same argument,

$$d_- := \min\left\{\int_{I'_+} f_+\,du, \int_{I'_-} f_-\,du\right\} > 0, \qquad C_K(w) + P_- \geq d_-.$$

Finally $E_m - A > 0$, so both improvements remain strict after adding the same respective nonlinear errors. □

The right-half-plane condition is material: a reference on the imaginary axis can have a kernel of one sign. The strict comparison just proved can be arbitrarily small and its improved interval may still contain zero. It establishes neither a uniform useful gain as height increases nor optimality among estimators retaining the full Laplace law. It compares two specified source relaxations, not all methods of arithmetic evaluation.

7.4 Conditional applicability and the remaining source issue

Proposition 7.5 (Local applicability at every putative positive double)**.** *Suppose the unchanged theta field has an ordinary double at $(0, a)$ with $a > 0$. On a sufficiently small fixed proper two-root window about that event, the unshifted reference (7.8) satisfies the hypotheses of Theorems 7.2 and 7.4 for every sufficiently large cutoff m. The assertion is pointwise in the event; it supplies no uniform cutoff over all heights and no contradiction to the assumed double.*

Proof. In the quantitative double-window construction, impose additionally $R \leq a/2$ and then choose the inner radius and time widths with the same inequalities. The proof of its positive wall margins is unchanged. Every point in the spatial disk has positive real part. On this fixed compact window the complete theta tails tend uniformly to zero on both nested walls. For all sufficiently large m, their absolute bounds are less than half the corresponding wall floors of H. Rouché's theorem gives two zeros of $F = H - E$ inside each disk, none in the collar, and an outer reference floor at least half that of H. In particular the reference roots lie in the right half-plane, and $|E/F| \leq |E|/(|H| - |E|) < 1$ on the outer wall.

The explicit sufficient ratio (7.12) also tends to zero. By (7.5),

$$E_m \int_0^\infty W(u)\cosh(Ru)\,du \leq \frac{E_m}{A}\int_0^\infty R_m(u)e^{tu^2}\cosh(Ru)\,du.$$

The mass ratio is bounded by (7.7); the complete tail integral tends to zero uniformly on the compact time interval by dominated convergence. The reference floor stays bounded below, so (7.12) eventually holds. The cited theorems now apply. □

The reference in (7.8) is essential. A shifted or reflection-corrected head changes the residual; the corresponding connector terms and shifted integration rays must then be restored before applying a source identity. The present proof uses the unshifted head and its complete tail, and makes no claim to resolve their small signed cancellation. This distinction is developed in [14].

There is also a limitation within the weighted-square source class. Positive weights for which the relevant source sums and integrals converge leave the same construction with masses a_n, b_n multiplied by those weights. Thus positivity of the Laplace law does not characterize the canonical coefficient sequence. When another source agrees through mode $m+1$ and has the same source value at zero, it has the same F, W, A, E_m and hence the same monotone response enclosure at any fixed admissible reference window. Its actual probability law and response need not agree with theta's. No assertion that a prescribed window contains a collision of that other source follows from such a match. This observation is separate from the finite envelope-moment control in Theorem 6.1; their preservation hypotheses are different.

The analytic output is therefore the actual-source enclosure (7.13), its strict matched comparison (7.16), and its conditional applicability near each putative ordinary double. The sign of the full theta discriminant lower bound remains open. The proof and control boundaries were internally checked in [12]; these statements do not claim independent specialist validation.

8 Discriminant evolution for a finite cluster of arbitrary size

The two-root equation has a regular extension to every finite cluster, but its constant forcing does not. This distinction gives both a useful general identity and exact limits on what can be inferred from two real cap signs. The results in this section develop the analytic statements recorded in [11]; their earlier internal review is [12].

8.1 A regular equation through collisions

Let D be a bounded Jordan complex-time domain with piecewise smooth boundary, and let H be holomorphic near $\overline{D} \times \{z : |z-a| \le R\}$, with $H_t = -H_{zz}$. Assume that, for every $t \in \overline{D}$, $H(t,\cdot)$ has exactly $m \ge 2$ zeros in the outer disk, all in $|z-a| < r < R$, and has no zero on the outer circle. These assumptions are understood with strict margins sufficient to extend them to a neighborhood of the closed product. The cluster factorization gives

$$P(t,z) = \prod_{i=1}^{m} (z - z_i(t)), \qquad H(t,z) = U(t,z)P(t,z), \qquad q = \frac{U_z}{U}, \qquad f = \frac{H_z}{H}, \tag{8.1}$$

where P has single-valued holomorphic coefficients and U is holomorphic and nonvanishing on the spatial disk. Individual root labels need not be single-valued. In particular, q includes the complete exterior contribution, including any zero-free factor of H.

Put $N = m(m-1)/2$ and define

$$\mathfrak{D} = \prod_{i<j} (z_i - z_j)^2, \qquad \mathfrak{T} = \sum_{i<j} \prod_{\substack{k<\ell \\ (k,\ell)\neq(i,j)}} (z_k - z_\ell)^2. \tag{8.2}$$

The empty product is one. Both expressions are symmetric polynomials in the roots and hence are holomorphic functions of the coefficients of P. Thus (8.2) defines $\mathfrak{T}$ at collisions; the quotient expression $\mathfrak{T} = \mathfrak{D} \sum_{i<j} (z_i - z_j)^{-2}$ is used only when the roots are distinct.

Theorem 8.1 (Regular finite-cluster discriminant equation)**.** *Under these assumptions,*

$$\mathfrak{D}' = 8\mathfrak{T} + B\mathfrak{D}, \tag{8.3}$$

$$B(t) = \frac{2}{2\pi i}\oint_{|\zeta-a|=R} \frac{H_z(t,\zeta)}{H(t,\zeta)}\frac{P_{zz}(t,\zeta)}{P(t,\zeta)}\,d\zeta. \tag{8.4}$$

The coefficient B is holomorphic through every internal collision. For any holomorphic scalar $\kappa(t)$, replacing H_z by $H_z - \kappa H$ in (8.4) leaves it unchanged. If, on the complete outer wall,

$$|H| \geq \delta > 0, \qquad |H_z - \kappa H| \leq L_\kappa, \tag{8.5}$$

then

$$|B| \leq \frac{2R\,m(m-1)L_\kappa}{\delta(R-r)^2}. \tag{8.6}$$

In particular, the bound contains no inverse internal root separation.

Proof. At a simple zero, differentiation and the heat equation give

$$z_i' = \frac{H_{zz}(t,z_i)}{H_z(t,z_i)} = 2\sum_{j\neq i}\frac{1}{z_i - z_j} + 2q(t,z_i). \tag{8.7}$$

Writing $d_{ij} = z_i - z_j$, we obtain

$$\frac{z_i' - z_j'}{d_{ij}} = \frac{4}{d_{ij}^2} - 2\sum_{k\neq i,j}\frac{1}{(z_i - z_k)(z_j - z_k)} + 2\frac{q(t,z_i) - q(t,z_j)}{d_{ij}}. \tag{8.8}$$

The middle terms cancel after summation over all pairs, because every unordered triple contributes a multiple of the identity

$$\frac{1}{(x-z)(y-z)} + \frac{1}{(x-y)(z-y)} + \frac{1}{(y-x)(z-x)} = 0.$$

Logarithmic differentiation therefore yields

$$\frac{\mathfrak{D}'}{\mathfrak{D}} = 8\sum_{i<j}\frac{1}{d_{ij}^2} + 4\sum_{i<j}\frac{q(t,z_i) - q(t,z_j)}{z_i - z_j}. \tag{8.9}$$

The divided differences in the last term extend holomorphically when their arguments coincide. Cauchy's formula and

$$\frac{P_{zz}(t,\zeta)}{P(t,\zeta)} = 2\sum_{i<j}\frac{1}{(\zeta - z_i)(\zeta - z_j)}$$

give

$$4\sum_{i<j}\frac{q(t,z_i) - q(t,z_j)}{z_i - z_j} = \frac{2}{2\pi i}\oint_{|\zeta-a|=R} q(t,\zeta)\frac{P_{zz}(t,\zeta)}{P(t,\zeta)}\,d\zeta.$$

Replacing q by $f - P_z/P$ introduces the rational integral of P_zP_{zz}/P^2. All finite poles of this rational function lie inside the contour, and it is $O(\zeta^{-3})$ at infinity. Its total residue is consequently zero. This proves (8.4). The same reasoning applied to $P_{zz}/P = O(\zeta^{-2})$ proves the exact scalar subtraction.

Every denominator in (8.4) is nonzero on the integration circle, so the coefficient is holomorphic in time. The simple-root set is dense: at any time, the local heat splitting of a zero of multiplicity k

has distinct Hermite leading roots after a nonzero small time increment. The finitely many clusters in that fiber can all be separated by one sufficiently small increment. Thus $\mathfrak{D}$ is not identically zero. Multiplying (8.9) by $\mathfrak{D}$ proves (8.3) on a dense set, and then everywhere by holomorphic continuation.

Finally, $|\zeta - z_i| \geq R - r$ on the circle, whence

$$\left|\frac{P_{zz}}{P}\right| \leq \frac{m(m-1)}{(R-r)^2}.$$

Using the contour length $2\pi R$ and (8.5) in (8.4) proves (8.6). □

For $m = 2$ one has $\mathfrak{T} = 1$, so the theorem is exactly the two-root evolution equation. For larger clusters, it keeps every root in the birth while retaining the same power two of the spatial collar in the coefficient bound.

8.2 An explicit window at a hypothesized full birth

For the complete theta source, write

$$B_j = M_j(T_0, R_0) = \int_0^\infty \Phi(u) u^j e^{T_0 u^2} \cosh(R_0 u)\, du, \qquad T_0, R_0 > 0.$$

These finite positive moments bound all derivatives needed below.

Proposition 8.2 (Parameterized geometry at multiplicity m)**.** *Suppose $a \in \mathbb{R}$ and $H(0,\cdot)$ has exact multiplicity $m \geq 2$ at a. Set*

$$q_m = \frac{|\partial_z^m H(0,a)|}{m!} > 0, \qquad R = \min\left\{\frac{R_0}{2}, \frac{(m+1)! q_m}{4B_{m+1}}\right\}, \qquad r = \frac{R}{2}, \tag{8.10}$$

$$h = \min\left\{\frac{T_0}{2}, \frac{q_m r^m}{4\sqrt{2}B_2}\right\}. \tag{8.11}$$

For $|\operatorname{Re} t|, |\operatorname{Im} t| \leq h$, each of the two disks of radii r and R contains exactly m roots, and

$$|H(t,z)| \geq \frac{q_m s^m}{2} \quad \text{if } |z-a| = s, \quad s \in \{r, R\}. \tag{8.12}$$

The complete coefficient in (8.4) obeys

$$|B| \leq \frac{16m(m-1)B_1}{q_m R^{m+1}}. \tag{8.13}$$

Proof. Put $c_m = \partial_z^m H(0,a)/m!$. Spatial Taylor expansion, with the complete derivative bound on the segment from a to z, gives

$$|H(0,z) - c_m(z-a)^m| \leq \frac{B_{m+1} s^{m+1}}{(m+1)!} \leq \frac{q_m s^m}{4} \qquad (|z-a| = s \in \{r, R\}).$$

Along the straight time segment, $|H_t| = |H_{zz}| \leq B_2$ and $|t| \leq \sqrt{2}h$, so

$$|H(t,z) - H(0,z)| \leq \sqrt{2} h B_2 \leq \frac{q_m r^m}{4}.$$

The total error is at most $q_m s^m/2$, strictly less than $|c_m(z-a)^m|$. Rouché's theorem gives both root counts and the triangle inequality gives (8.12). The inequalities have strict slack relative to the zero-avoidance and root-count requirements, so the contract extends near the closed product. In (8.6), take $\kappa = 0$, $L_\kappa = B_1$, $\delta = q_m R^m/2$ and $R - r = R/2$, obtaining (8.13). □

This construction is conditional on the event and its nonzero order-m derivative. It gives a positive window at every such finite event, with all theta modes retained in B_j. It supplies neither a sign of the discriminant nor nonvanishing on the complex time boundary. The dependence on q_m and m is explicit and may deteriorate with height.

8.3 Birth order, forcing, and charge

Suppose the entire cluster coalesces at (t_0, z_0). Write $s = t - t_0$ and $s = \rho^2$. The local heat splitting is

$$z_i(t_0 + \rho^2) = z_0 + \sqrt{2}\lambda_i \rho + O(\rho^2), \tag{8.14}$$

where $\lambda_1, \dots, \lambda_m$ are the distinct real roots of the monic probabilists' Hermite polynomial He_m. Indeed, the leading part of parabolic weight m is

$$e^{-s\partial_z^2}(z - z_0)^m = (2s)^{m/2}\,\mathrm{He}_m\left(\frac{z - z_0}{\sqrt{2s}}\right).$$

All higher initial spatial Taylor terms have larger parabolic weight; analytic continuation of the simple scaled roots gives (8.14).

Corollary 8.3 (Order at a full birth)**.** *At a full multiplicity-m birth,*

$$\mathfrak{D}(t) = d_m s^N + O(s^{N+1}), \qquad d_m = 2^N \prod_{i<j} (\lambda_i - \lambda_j)^2 > 0, \tag{8.15}$$

$$\mathfrak{T}(t) = \frac{N d_m}{8} s^{N-1} + O(s^N). \tag{8.16}$$

If $\mathfrak{D}$ is nonzero on ∂D, the product-domain charge is

$$Q = \frac{1}{2\pi i} \oint_{\partial D} \frac{\mathfrak{D}'}{\mathfrak{D}}\, dt = \frac{8}{2\pi i} \oint_{\partial D} \frac{\mathfrak{T}}{\mathfrak{D}}\, dt. \tag{8.17}$$

Proof. Taking the product of squared differences in (8.14) gives the leading coefficient in (8.15). Holomorphy in s removes any noninteger power from its remainder. Substitution in (8.3), using holomorphy of B, proves (8.16). The first equality in (8.17) is the finite-cluster charge formula. For the second, integrate (8.3) divided by $\mathfrak{D}$; the integral of the holomorphic coefficient B is zero. □

The exterior contribution still changes $\mathfrak{D}$, $\mathfrak{T}$ and their zero locations through (8.3); its vanishing contour integral is not permission to omit it from the evolution. For $m > 2$ the forcing vanishes at a full birth. This removes the constant forcing used by the two-root univalence argument. A full birth has charge N, while two real discriminant signs record at most its parity. Conjugate nonreal event pairs can contribute further even charge without changing those signs.

8.4 A cubic real-time monotonicity theorem

Assume now that a is real, D is invariant under conjugation and $H(\bar{t}, \bar{z}) = \overline{H(t, z)}$. For $m = 3$, depress the cluster polynomial:

$$P(t, z) = (z - c(t))^3 + p(t)(z - c(t)) + q_0(t).$$

The notation q_0 distinguishes the constant coefficient from U_z/U. Direct expansion gives

$$\mathfrak{D} = -4p^3 - 27q_0^2, \qquad \mathfrak{T} = 9p^2. \tag{8.18}$$

For the second identity, $\mathfrak{T}$ is a translation-invariant symmetric polynomial of degree four. After centering, the fundamental symmetric polynomials have degrees two and three, so it is a constant multiple of p^2. Substituting roots $(-1, 0, 1)$ gives the constant 9.

Proposition 8.4 (At most one real cubic collision time)**.** *On each real interval contained in D, a proper real-symmetric three-root heat cluster has at most one collision time. An independently located collision at a strictly negative time in such an interval excludes an additional collision at time zero in that interval.*

Proof. Choose an antiderivative of B which is real on the real interval, and put $\mu = \exp(-\int B)$. Then $\mu > 0$ there and (8.3)–(8.18) imply

$$(\mu\mathfrak{D})' = 72\mu p^2 \geq 0. \tag{8.19}$$

If p is not identically zero, its real zeros are isolated. The integral of the right side over every nontrivial subinterval is then positive, so $\mu\mathfrak{D}$ is strictly increasing and has at most one zero. If p is identically zero, $\mu\mathfrak{D}$ is constant and nonzero, since $\mathfrak{D}$ cannot be identically zero. There are no collisions in that case. □

A real event in this cubic cluster can be a double, of charge one, or a triple, of charge three. Thus the proposition bounds the number of real event locations, not their total charge. It also gives no bound on nonreal event times. At complex time p^2 need not have positive real part, even if μ lies in a fixed open half-plane. For $m \geq 4$, even the real inequality in (8.19) need not persist.

8.5 Exact controls for the failure of the cap formula

The following polynomial heat solutions pay the cluster and boundary conditions explicitly. They disprove the extension of the two-root cap formula to arbitrary multiplicity, without challenging its two-root hypothesis. Their conclusions are algebraic; the exact arithmetic replay in [19] is supplementary support.

Proposition 8.5 (A cubic with three event times)**.** *Let*

$$H(t, z) = z^3 - 6tz + 2, \qquad D = \{t : |\operatorname{Re} t| < 1,\ |\operatorname{Im} t| < 1\}, \qquad r = 4, \quad R = 5, \quad a = 0.$$

This is a proper three-root heat cluster with $B = 0$. Its total charge is three, although $(\operatorname{sgn}\mathfrak{D}(1) - \operatorname{sgn}\mathfrak{D}(-1))/2 = 1$.

Proof. One has $H_t = -6z = -H_{zz}$. On the inner wall and closed time rectangle,

$$|6tz - 2| \leq 24\sqrt{2} + 2 < 64 = |z^3|.$$

Rouché's theorem puts all three polynomial roots strictly in the inner disk, hence also in the outer disk; both walls are nonzero. Here $U = 1$, and thus $B = 0$. Formula (8.18) gives

$$\mathfrak{D}(t) = 864t^3 - 108, \qquad \mathfrak{T}(t) = 324t^2.$$

The event times are the three simple roots of $t^3 = 1/8$:

$$\frac{1}{2}, \qquad -\frac{1}{4} + \frac{\sqrt{3}}{4}i, \qquad -\frac{1}{4} - \frac{\sqrt{3}}{4}i.$$

At each event the repeated root is $z = 1/(2t)$, of modulus one; $H_{zz} = 6z \neq 0$ there, so each event is an ordinary double. All three lie strictly within the product domain, and no event is on its boundary. Each has charge one. Finally, $\mathfrak{D}(-1) = -972$ and $\mathfrak{D}(1) = 756$, proving the claimed discrepancy. □

Proposition 8.6 (A quadruple endpoint with equal positive cap signs)**.** *On the same time rectangle, let*

$$H(t,z) = z^4 - 12tz^2 + 12t^2, \qquad a = 0, \quad r = 5, \quad R = 6.$$

This proper four-root heat cluster has $B = 0$, nonzero time-boundary discriminant, equal positive real cap discriminants and total charge six. For every positive real time its spatial roots are real and simple.

Proof. The heat equation follows by direct differentiation. The inner-wall estimate is

$$|12tz^2 - 12t^2| \le 300\sqrt{2} + 24 < 625 = |z^4|,$$

so all four roots lie strictly in the inner disk throughout the closed rectangle. Again $U = 1$ and $B = 0$. Set $\alpha = 6 + 2\sqrt{6}$, $\beta = 6 - 2\sqrt{6}$. The factorization

$$H(t,z) = (z^2 - \alpha t)(z^2 - \beta t)$$

gives roots $\pm\sqrt{\alpha t}$ and $\pm\sqrt{\beta t}$. The squared differences within the two opposite pairs contribute $16\alpha\beta t^2$, and the four cross-pair squared differences contribute $(\alpha - \beta)^4 t^4$. Therefore

$$\mathfrak{D}(t) = 16\alpha\beta(\alpha - \beta)^4 t^6 = 1{,}769{,}472\, t^6, \qquad \mathfrak{T}(t) = 1{,}327{,}104\, t^5,$$

where the last identity also follows from $\mathfrak{D}' = 8\mathfrak{T}$. The only event is $(0,0)$, of charge six. Both cap discriminants are positive, and no point of the time boundary is an event. Since $\alpha > \beta > 0$, the four displayed roots are real and distinct for $t > 0$. Notice also that $\mathfrak{T}$ changes sign on real time. □

More generally, the Hermite family $e^{-t\partial_z^2} z^m$ has $\mathfrak{D} = d_m t^N$, $\mathfrak{T} = N d_m t^{N-1}/8$ and $B = 0$; a sufficiently large spatial disk gives a proper finite window. Thus zero exterior phase alone is compatible with every full-birth charge. The controls above concern the specified phase-and-cap inference. They do not preclude other boundary estimates or estimates using additional properties of the theta source.

8.6 Why centered variance cannot replace the discriminant

Define

$$c = \frac{1}{m}\sum_i z_i, \qquad V = \sum_i (z_i - c)^2 = \frac{1}{m}\sum_{i<j} (z_i - z_j)^2.$$

The equality follows by expanding the pairwise sum. This quantity is holomorphic even when root labels are not.

Proposition 8.7 (Centered second-moment evolution)**.** *For a proper m-root heat cluster,*

$$V' = 2m(m-1) + 4\sum_i (z_i - c) q(t, z_i). \tag{8.20}$$

At a full birth (t_0, z_0),

$$V = 2m(m-1)(t - t_0) + O((t - t_0)^2). \tag{8.21}$$

Proof. Since $\sum_i (z_i - c) = 0$, differentiating gives $V' = 2\sum_i (z_i - c) z_i'$. Substitute (8.7). For each unordered internal pair the contribution is

$$4\left(\frac{z_i - c}{z_i - z_j} + \frac{z_j - c}{z_j - z_i}\right) = 4.$$

There are N pairs, proving (8.20) at simple fibers. The remaining symmetric sum is holomorphic, for it equals

$$\frac{1}{2\pi i}\oint_{|\zeta-a|=R}(\zeta-c)q(t,\zeta)\frac{P_z(t,\zeta)}{P(t,\zeta)}\,d\zeta.$$

Continuation proves the identity everywhere. At a full birth $V=0$ and each $z_i-c=0$, so evaluating (8.20) gives (8.21). □

The real sign transition in (8.21) is compatible with every Hermite birth. Moreover, V need not have the discriminant's zero set: the cubic in Proposition 8.5 has $V=12t$, hence $V(0)=0$ but $\mathfrak{D}(0)=-108\neq 0$. On an all-real fiber, $V=0$ does force full coalescence, but that fiber condition cannot be assumed on an unknown negative cap. Conversely, $V>0$ does not exclude a partial collision in a larger real cluster. Neither centered variance nor two discriminant signs therefore replace the full multiplicity-sensitive event calculation.

9 Supporting certificates and exact arithmetic checks

The analytic identities and counterexamples above do not depend on the computations in this section. The numerical work serves two purposes: to demonstrate that the complete theta source can satisfy the finite boundary hypotheses, and to check implementations against exact models. All reported numerical claims retain the scope and internal-review status of their recorded calculations.

9.1 A complete-theta application of the two-root theorem

The interval calculation in [8, 18] uses

$$|\operatorname{Re} t|,|\operatorname{Im} t|\le 10^{-2},\qquad a=\frac{71}{2},\quad r=8,\quad R=10. \tag{9.1}$$

Both circular root counts are two. The enclosing data, rounded outward for display, are

Quantity	Certified bound
Inner circular wall floor	$> 8.36937\times 10^{-7}$
Outer circular wall floor	$> 7.57583\times 10^{-7}$
Exterior coefficient modulus	$\lvert A\rvert < 3.489078$
Phase product	$10^{-2}\sup\lvert A\rvert < 0.034891$
Discriminant at each real time cap	$\Delta(\pm 10^{-2}) > 144$
Complete source integrals	4146

The two-root cap theorem therefore gives $Q=0$ on this window. No RH assumption or tabulated zero location is an input.

The certificate covers entire arcs and the full complex-time rectangle, not only sampled points. It uses 256 arcs on each circle, complete jets of orders zero through fifteen at the time-zero anchors, and a joint Taylor expansion of total degree seven. Complete moments pay the order-eight remainder. All omitted source modes and exterior integration tails are bounded. Zero-avoiding convex disks around the boundary image arcs justify the polygonal winding counts. The coefficient estimate retains the coupled numerator $H_z+(3/10)H$ before taking absolute values. If d_k and n_k are the full-arc lower and upper bounds for the denominator and this numerator, respectively, the calculation uses

$$|A|\le\frac{4R}{(R-r)^2}\frac{1}{256}\sum_{k=0}^{255}\frac{n_k}{d_k}.$$

At each real time cap, the certified signs at spatial points $28, 29, 41, 43$ are $+, -, -, +$. The intermediate value theorem and the total count of two put one simple root in each of $(28, 29)$ and $(41, 43)$. Their gap exceeds twelve, proving the displayed discriminant bound. Thus the cap signs have an independent real-source payment.

This is a separated-pair example. Its large cap gap does not address an unknown shrinking gap near a hypothetical endpoint. The same boundary and phase procedure also admits the exact quadratic collision $H = (z - 71/2)^2 - 2t$, returning charge one from its opposite cap signs. It therefore does not silently assume that events are absent.

9.2 Earlier direct comparisons

The earlier finite experiments [3, 5] compared specified boundary and direct joint-exclusion procedures using complete source data. Their certified margins and acquisition counts were

Experiment	Boundary margin	Direct margin	Source integrals
One-zero ellipsoid	$> 7.495 \times 10^{-5}$	$> 7.993 \times 10^{-5}$	8 versus 6
Two-zero thin box	$> 3.5683 \times 10^{-7}$	$> 3.9348 \times 10^{-7}$	1032 each

The thin box has $|\operatorname{Re} t|, |\operatorname{Im} t| \le 0.01$, $28 \le \operatorname{Re} z \le 43$, and $|\operatorname{Im} z| \le 0.01$. Both methods used the same source acquisition and 2056 boundary vertices. The direct procedure had the stronger certified margin in these declared implementations. This is neither a universal complexity ordering nor evidence against other boundary geometries. The analytic reference-transfer result in Appendix A explains why an already certified holomorphic reference does not by itself confer an improved uniform margin on the boundary method.

9.3 Exact controls and reproducibility

The standard-library rational calculation [19] checks the Hermite discriminants for $2 \le m \le 8$, five finite clusters with nonzero exterior forcing, the cubic expression for T, and quadratic source response. It also verifies the explicit boundary-paid affine homotopy example from the signed-response section. These checks agree with the proofs, including the negative-time signs of the many-root forcing. They are finite algebra checks, not a proof for all m.

The earlier piecewise affine boundary-degree controls [4] return degrees $+1, -1, +1, 0$ for, respectively, a complex-oriented linear map, an orientation-reversing real map, an enclosed quadratic heat event, and a quadratic heat field with its event outside the box. The negative-degree example deliberately fails holomorphicity. It checks the orientation convention rather than holomorphic positivity.

The repository records supply scripts, arithmetic versions, parameter choices, source-tail and interpolation budgets, and small execution receipts. The theta runs use `python-flint 0.9.0`; the exact algebra checks require only Python's standard library. Source-array hashes bind the recorded computations but do not replace their analytic error estimates. No additional theta numerical region was acquired for this manuscript.

10 What remains to obtain endpoint exclusion

The geometric difficulty at an isolated hypothetical double has been resolved in a pointwise quantitative sense: a positive radius, collar, time width, pair of root counts, and phase margin follow from the nonzero second jet and complete moments. The resulting window correctly has charge one

when a double is assumed. A new exclusion must therefore use information beyond those universal local properties.

A sign for the unchanged theta source. The signed response and the canonical-tail mixture retain correlations that absolute moment bounds discard. Their remaining task is to prove, with a declared reference and the entire residual included, either a strictly positive incoming discriminant or a strictly negative event location. For a proper two-root window with a positive outgoing cap, the first gives charge zero. Alternatively, an independently located negative-time event uses the unique event allowed by the phase theorem and excludes another event at zero. Neither required theta-specific inequality has been established here. A strict improvement over another enclosure can still leave zero inside the resulting interval.

Height coverage. The windows exist separately at every finite hypothetical double height. Their sizes depend on the local curvature; the displayed construction can shrink with height. The source sign, error terms, tail cutoff and event-location margin must therefore be controlled with the same dependence. A calculation on one fixed interval, or an unquantified limiting statement about the head cutoff, cannot discharge the universal quantifier over endpoint heights. No common negative all-real time slice is being proposed.

Every multiplicity. The all-root identity $D' = 8T + BD$ retains the whole cluster and its exterior contribution. For $m > 2$, the forcing T is no longer constant. The cubic control shows that nonreal time events survive even when $B = 0$; the quartic control has charge six despite equal positive real cap signs. Applying the two-root theorem to an arbitrarily selected pair at a higher birth is invalid, because the other birth roots destroy the required isolating collar. The cubic real-event monotonicity provides a possible conditional event-location argument, but no complete-source location estimate is supplied. Higher multiplicities still require an exclusion compatible with their full discriminant.

The finite-moment control constructed above has exactly its stated preservation properties; it does not realize the actual theta moment vector. Broader collision controls in the project have different failures of the theta premises: Euler-factor controls change arithmetic data; finite square-source and finite-prime controls lack the complete normalized coefficient/reflection conjunction; reflected or smooth positive-source controls lack the canonical square-source expansion; the close positive endpoint controls in [17] fail ordinary Dirichlet-series arithmetic. Their preservation lists must not be combined. They rule out arguments using only their shared premises, not estimates specific to the unchanged complete theta source.

In the task register, the hypothetical-double window is complete in its stated parameterized scope (ECE-01). The distinguishing sign (ECE-02), its all-height use (the signed part of ECE-04), and higher-multiplicity exclusion (the remaining part of ECE-05) are open. Endpoint exclusion and RH-conditional simplicity (ECE-03 and ECE-06) remain contingent on those results. The manuscript is an analytic consolidation, not an upgrade of those statuses.

Acknowledgements and research provenance

This manuscript was drafted for Edward Baker with substantial assistance from OpenAI's Codex language-model system. The manuscript drafting and internal review session recorded model `gpt-6-astra`, reasoning effort `ultra`. The underlying 21 October 2026 notes record their own configurations, including earlier `max` sessions; those provenance distinctions are retained in the source records. Separate agents of the same model assisted in drafting and checking sections. These

checks do not constitute independent specialist review or formal verification. The author designation above identifies the intended recipient of the draft and does not imply that he has approved every claim. The investigation's notes, reviews, progress register and chat archive preserve the detailed research history; no manuscript snapshot series is created.

A Holomorphic references and boundary interpolation

This appendix makes explicit two analytic ingredients of the earlier finite comparisons [4]. They are separate from the signed source estimates used in the endpoint program.

Proposition A.1 (Minimum norm for a zero-free square holomorphic map)**.** *Let $\Omega \Subset \mathbb{C}^n$ be a bounded domain with piecewise smooth boundary, and let F be holomorphic near $\overline{\Omega}$, with values in $\mathbb{C}^n$. Suppose $b = \min_{\partial\Omega} \|F\| > 0$. If $\deg(F, \Omega, 0) = 0$, then F has no zero in $\overline{\Omega}$ and*

$$\min_{\overline{\Omega}} \|F\| = b.$$

Proof. For $\|v\| < b$, the fiber $F^{-1}(v) \cap \Omega$ is a compact analytic subset of the open Euclidean domain: it cannot approach the boundary. Such a subset is finite. Indeed, on a compact irreducible analytic component the maximum principle makes every holomorphic coordinate function constant, excluding positive dimension; compactness then makes the discrete set finite. At each isolated holomorphic preimage the local degree is a positive integer, the analytic multiplicity. The degree at v is their sum. It is constant for $\|v\| < b$, since that ball misses $F(\partial\Omega)$. Degree zero at 0 therefore implies that every such fiber is empty. This proves the claimed interior lower bound, and compactness of the boundary gives equality. □

Corollary A.2 (Transfer of the same certified margin)**.** *Under the hypotheses of Proposition A.1, let G be another square holomorphic map near $\overline{\Omega}$ and suppose $\sup_{\partial\Omega} \|G - F\| \le e < b$. Then*

$$\|G\| \ge b - e \quad \text{on } \overline{\Omega}, \qquad \deg(G, \Omega, 0) = 0.$$

Proof. Apply the scalar maximum-modulus principle to every unit linear functional of $G - F$. Taking the supremum over those functionals gives $\sup_{\overline{\Omega}} \|G - F\| \le e$. The norm inequality follows from Proposition A.1; the straight boundary homotopy $F + s(G - F)$ is nonzero and preserves degree. □

The same supplied reference-degree proof, boundary floor and error certificate thus yield a direct interior joint margin without additional source acquisition. This is a comparison of those data, not a lower bound for the cost of every boundary algorithm. Computing the reference degree has a cost shared by both uses.

The margin is sharp even for a heat jet requiring both components. On $|\operatorname{Re} t|, |\operatorname{Im} t| < 1/10$ and $|\operatorname{Re} z|, |\operatorname{Im} z| < 3/2$, set

$$F = (z^2 - 1 - 2t, 2z), \qquad G = (z^2 - 1 - 2t + \eta, 2z), \quad 0 < \eta < 4/5.$$

Writing $s = |z|^2$ and $k = 1 + 2t$ gives

$$\|F\|^2 \ge (s - |k|)^2 + 4s = |k|^2 + (4 - 2|k|)s + s^2 \ge 16/25,$$

because $\operatorname{Re} k \ge 4/5$ and $|k| < 2$. Equality holds at $t = -1/10, z = 0$. The identical argument with $k - \eta$ gives exact floor $4/5 - \eta$ for G, while $\|G - F\| = \eta$. At time zero, the first component of F vanishes at $z = \pm 1$, and its second component at $z = 0$. Neither component by itself supplies the joint floor.

A.1 A complete interpolation payment

Suppose a convex boundary simplex is approximated by affine interpolation of the jets $H_j = \partial_z^j H$, $j = 0, 1$. Let (t_i, z_i) be its vertices, λ_i barycentric weights, and $(t, z) = \sum_i \lambda_i (t_i, z_i)$. Put

$$V_t = \sum_i \lambda_i |t_i - t|^2, \qquad V_z = \sum_i \lambda_i |z_i - z|^2.$$

If complete moments $M_k(T, R)$ bound all segment derivatives, Taylor's formula and cancellation of the linear barycentric term give

$$\left| \sum_i \lambda_i H_j(t_i, z_i) - H_j(t, z) \right| \le I_j := \frac{1}{2} \left(M_{j+4} V_t + 2 M_{j+3} \sqrt{V_t V_z} + M_{j+2} V_z \right). \tag{A.1}$$

To check this, the second directional derivative has terms $H_{j,tt} = H_{j+4}$, $H_{j,tz} = -H_{j+3}$ and $H_{j,zz} = H_{j+2}$. Integrating the Taylor remainder contributes $1/2$, and Cauchy–Schwarz bounds the mixed barycentric sum. For a real coordinate with range length h, its variance is at most $h^2/4$: this follows by taking expectations in $(X - a)(b - X) \ge 0$ and maximizing $(\mathbb{E}X - a)(b - \mathbb{E}X)$. Summing over the two time and two space coordinates bounds V_t, V_z explicitly.

If the vertex errors in H_j are at most δ_j, convexity keeps the interpolated acquisition errors at most δ_j. Thus the full Euclidean jet error on the simplex is at most

$$e_\sigma = \sqrt{(\delta_0 + I_0)^2 + (\delta_1 + I_1)^2}. \tag{A.2}$$

A certified lower bound b_σ for the norm on the entire affine image, with $e_\sigma < b_\sigma$, gives a nonzero straight boundary homotopy. Vertex norms alone are insufficient for b_σ; for example, enclosing the image convex hull in a coordinate box permits a lower bound by that box's distance from zero.

For completeness, a finite piecewise affine degree computation can cone a coherently triangulated box boundary to its center, assigning the artificial target value zero to the cone vertex. This is an extension of the boundary map, not an interior approximation to G. For each four-simplex form the domain and target matrices $X = [x_0\ x_1\ x_2\ x_3]$ and $Y = [y_0\ y_1\ y_2\ y_3]$ relative to the cone center. A sufficiently small regular target v contributes $\operatorname{sgn}(\det Y / \det X)$ exactly when $Y^{-1} v$ has positive coordinates with sum less than one. Choose $\|v\| < \min_\sigma b_\sigma$ and verify that it is on no simplex image face and has no preimage in a singular simplex. The sum is the boundary degree, by additivity and homotopy invariance. Rational linear algebra can verify these finite conditions exactly. The artificial center uses no evaluation of the actual heat field. Finally, holomorphic positivity is what makes zero degree exclude a zero of (H, H_z); that last conclusion would fail for a general real map.

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